From Lecture 13 – Slide 18

Homework Solution: `int max(int x, int y)`

in C:

```
int max(int x, int y) {
   int result = x;
   if (y > x)
      result = y;
   return result;
}
```

version 1 – with **jx** instruction - **mine**

in assembly:

```
max:
    # x in %edi, y in %esi, result in %eax
    movl %edi, %eax   # result = x
    cmpl %edi, %esi   # if y <= x then
    jle   endif       # return x (result)
    movl %esi, %eax   # otherwise return y
endif:
    ret
```

We branch (jump) when the condition (`y > x`) is false, i.e., when (`y <= x`)

-> This technique is called *"coding the false condition first"*

or *"taking care of the false condition first"*

Example: alternate `int max(int x, int y)`

version 2 - with **`jX`** instruction – **from gcc (-Og)**

in C:

```
int max(int x, int y) {
  int result = x;
  if (y > x)
    result = y;
  return result;
}
```

in assembly:

```
max:
    # x in %edi, y in %esi, result in %eax
    movl %esi, %eax # result = y
    cmpl %esi, %edi # if x < y then
    jl   endif      # return y (result)
    movl %edi, %eax # otherwise return x
endif:
    ret
```